

Taka Khoo

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takahoo.com |  | 

EXPERIENCE

Mozart AI (\$7M+ raised; Balderton-backed)

Mar 2026–Aug 2026

Founding Engineer, AI & Product, London

- Shipped TypeScript/React library, onboarding, and authenticated Liveblocks collaboration for a platform with **250K+ users** and **1M songs generated**; persisted track/mixer/MIDI state.
- Built multi-model audio evaluation: BPM/key/lyric checks, stem/semantic metrics, Bradley–Terry A/B ranking, PostHog telemetry, and a LiveKit voice-agent pipeline.

DARPA INGOTS (NARF Industries & Dartmouth LISP Lab)

Jun 2025–Jun 2026

AI Specialist / Cybersecurity Researcher

- Built AI-assisted Android vulnerability discovery, exploitability triage, verifier-gated PoCs, and structured evidence; developed cross-privilege exploit chains using Ghidra, ADB, and SELinux.
- Extended **ParseRE**'s QEMU binary labeling with an ELF corpus and grammar; built libelf/json-c/curl Docker harnesses and reproducible Cuttlefish evaluations.

Dartmouth Thayer School of Engineering

Jun 2025–Jun 2026

Head Teaching Assistant, Graduate Machine Learning

- Led two graduate ML courses; wrote **four textbooks** expanding Prof. Peter Chin's lectures, now used at Dartmouth.

Igloo Music

Feb–Jun 2024

Production & Mastering Engineer, Los Angeles

- Signal-chain patching, production, mastering, and **IMDb-credited dubbing** in a Grammy-winning studio.

SELECTED PRODUCTS, RESEARCH & THESES

MODULO, AI-Native Music Workstation

Independent product; M.S. thesis

- Built a C++/JUICE/Tracktion Mac workstation with VST/AU hosting, real-time audio, generative chords, audio-to-MIDI, and stem separation; shipped a **SwiftUI iOS companion to TestFlight** (StoreKit, Sign in with Apple, background audio).
- Designed the launch backend: **FastAPI + PostgreSQL** on Render and Supabase (RLS, migrations), **Stripe** subscriptions and token wallet, and a durable generation worker with MusicGPT/ElevenLabs adapters; live checkout, paid generation, and refund verified end to end.
- First-authored the revised UIST manuscript**: a within-subjects study halved time to satisfaction, tripled chord roots, nearly quadrupled chord qualities, and earned unanimous preference.

Lacquer, Automatic Restoration & Mastering

Paper draft; successor to my honors thesis

- Measure-first restoration of finished mixes: sparse declipping, cepstral echo removal (+16.4 dB SI-SDR), a fine-tuned band-split transformer for reverb (+5.9 dB where four released models gain at most 0.7 dB), and mastering held to norms from 103,838 released tracks. Thesis: 1.08B-parameter EnCodec-token U-Net with curriculum learning.

Unsplice, Speech Privacy in Federated ASR

First-author paper; code

- Showed one federated update leaks the client's speech features in closed form: recovered 98.5% of 1,417 LibriSpeech utterances (to 35 s by sequential decoding) with no transcript or optimisation; Whisper reads the audio at 3.8% WER.

Mikiri, Learned Search Stopping for Go

Paper; code

- Learned stopping rule and exact search memory on a frozen KataGo: **+228 Elo** (78.8% of 1,000 games) at equal mean visits, ahead of ten published stopping rules re-implemented on the same engine.

Audio Sliders, Controllable Text-to-Music

Paper; live demo; weights

- LoRA sliders on ACE-Step 1.5 and Stable Audio Open, scored by measured audio descriptors and music-quality models; newer axes discovered in 14,985 real recordings.

EDUCATION

Dartmouth College

Hanover, NH

M.S. Engineering Sciences, AI / Signal Processing / Music (*GPA: 3.92/4.00*)

Jun 2026

A.B., Engineering Sciences & Music, honors in both majors; music thesis album *My Life in Sound*

Jun 2025

SKILLS & INTERESTS

Engineering & AI: C++, Python, **Swift/SwiftUI**, TypeScript/JavaScript, React/Next.js, Node.js, FastAPI, SQL/PostgreSQL (Supabase, Neon, RLS), MongoDB, Docker, GCP, Render, Vercel, Cloudflare, Git/CI, LaTeX; PyTorch, TensorFlow, LLM agents, **Model Context Protocol (MCP)**, RAG, AI evaluation, audio/speech ML, DSP.

Product engineering: **0-to-1 full-stack systems**, iOS (StoreKit, TestFlight, Sign in with Apple), Stripe billing, real-time collaboration, API/data architecture, product analytics, A/B evaluation, rapid prototyping; Liveblocks, LiveKit, PostHog, JUICE, Tracktion Engine; **DAWs (expert)**.

Leadership: Co-founded Jump Ahead Tutoring (50+ tutors); coached 10+ Ivy League admits. **NCAA Division I swimmer**; broke Dartmouth school records twice and Hawaiian state records; perfect-pitch multi-instrumentalist.